

From Morphology to Phase

Dendritic Memory Encoding and PWD Readout in Self Aware Networks

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Abstract

A familiar object can be recognized even when it is seen from a new angle, under different lighting, or after part of it is hidden. The nervous system must therefore do more than count incoming spikes. It must preserve learned constraints that determine which spatial-temporal input patterns belong together and what a receiving cell should do when one of those patterns arrives. Established neuroscience already shows that dendritic branches are active nonlinear compartments, that branch excitability can store input features, that learning can cluster new spines, and that action-potential waveform can retain information about recent conductance history. The open problem is how those memory-bearing cellular conditions become a typed, receiver-relevant output rather than disappearing into a single firing-rate number.

This paper develops a measurable Self Aware Networks account of that transition. A neuron’s learned morphology, synapse placement, receptor and channel state, intrinsic excitability, intracellular organization, and recent activity jointly constrain which input patterns it recognizes. A compatible input evokes a multidimensional output whose phase, timing, duration, waveform, burst structure, amplitude, and release consequences can differ from the output expected under the receiver’s ongoing state. After the operation is established in ordinary physiological terms, this receiver-relative departure is named a **Phase Wave Differential (PWD)**. The proposal is not that anatomy travels down an axon. It is that learned anatomy and cellular state parameterize a transform, and the evoked PWD is a compressed functional readout of that transform.

We formalize the morphology-to-match-to-readout chain, provide a deterministic matched-rate demonstration, and specify causal experiments with branch intervention, typed-component ablation, and matched rescue. In the constructed demonstration, rate alone is uninformative while typed output and downstream response perfectly identify the two cellular states; removing every typed difference returns classification to chance. The result is an identifiability proof-of-concept, not a biological effect-size estimate. The empirical claim survives only if structural state predicts held-out output differences beyond input, recent history, and mean rate, and if selective perturbation and rescue show that identified output components alter identified receivers.

Keywords: dendritic computation; structural plasticity; neural coding; Phase Wave Differential; morphology; receiver-relative dynamics; Self Aware Networks

1. Begin With Recognition

Imagine touching the same piece of broccoli with two different parts of a whisker array, or seeing it after rotating it. The incoming pattern changes, yet the organism can still treat the object as the same kind of thing. A receiving neuron is not presented with a label saying *broccoli*. It receives distributed events at particular

branches, times, phases, and amplitudes while the cell is already in a changing electrical and biochemical state.

The practical problem is therefore:

How can a learned cellular structure recognize a relevant spatial-temporal relation and make that recognition available to the next part of the network?

A scalar artificial unit offers one useful approximation: multiply inputs by weights, sum them, and apply an activation function. A living neuron contains that kind of effective summation but also contains branch geometry, location-dependent nonlinearities, receptor and channel distributions, local inhibition, backpropagating events, adaptation, synaptic turnover, and a recent history that changes the effect of the next input. Reducing all of those conditions to one scalar can erase the very differences that made the pattern recognizable.

The proposed solution is straightforward before any new vocabulary is introduced:

1. learning changes a neuron's physical and functional organization;
2. that organization makes some input relations more effective than others;
3. a matching input produces a particular timed and shaped output;
4. the output is interpreted relative to the ongoing state of a receiver; and
5. downstream cells respond to the task-relevant part of the difference.

The remainder of the paper turns this plain-language operation into a formal, measurable hypothesis.

2. What Established Neuroscience Already Supplies

2.1 Dendrites are active computational structures

Dendrites are not merely passive wires. Their location-dependent conductances and regenerative events allow branches to respond nonlinearly to spatially and temporally organized input. Poirazi and Mel showed theoretically that nonlinear dendritic subunits and structural plasticity can greatly increase memory capacity compared with linear integration, particularly when learning can select where synaptic contacts are placed on dendritic subunits. Losonczy, Makara, and Magee then demonstrated branch-specific potentiation of dendritic spike-to-soma coupling, supporting the idea that branch excitability can store correlated input features. Smith and colleagues showed in vivo that visually evoked dendritic spikes contribute to orientation selectivity.

These results do not establish the full Self Aware Networks mechanism. They establish its necessary cellular premise: the physical arrangement and state of a dendritic tree can change the input-output function performed by a single neuron.

2.2 Learning can be spatially organized

Structural plasticity is not exhausted by a list of stronger or weaker synapses. Learning can alter spine formation and stabilization, and new spines can cluster on selected dendritic branches. Such clustering provides a plausible route by which repeated experience changes which input combinations can cooperate locally. The relevant unit of memory can therefore include the placement and interaction of contacts, not only the scalar efficacy assigned to each contact.

2.3 The output is richer than spike count

The all-or-none character of spike initiation does not imply that every emitted event is physiologically identical. De Polavieja and colleagues found that action-potential height and width in cortical pyramidal neurons reliably carried information about approximately 50 ms of preceding conductance history in their recordings, with more information than instantaneous firing rate alone. Spike timing, burst structure, adaptation, and transmitter release provide additional typed observables.

This matters because the question is not whether every output feature is always used. The question is whether a learned cellular state can leave a reproducible, causally effective signature in one or more output features after mean rate has been controlled.

3. The Historical SAN Genealogy

Micah Blumberg’s September 2012 source *Neurons are Programs transmitting their shape theory* argued that neuronal topology functions as a program and that an electrical packet can provide a sparse representation of the richer cellular condition that produced it. Companion 2012 records described potential difference, phase, amplitude, frequency, dendritic change, and whole-neuron state as parts of this readout problem. The phrase *transmitting shape* should be read functionally: a neuron does not send a miniature anatomical diagram down its axon. Its learned structure determines what it can recognize and how a match changes its output.

The public Git history fixes these records in 2022, while the source packet separately preserves the 2012 authored dates and contemporaneous context. The mature terms **Phase Wave Differential** and **Neural Array Projection Oscillation Tomography** appear in the 2022 SAN development sequence and must not be backdated to 2012. The source-faithful genealogy is therefore:

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2012: morphology/topology as program; sparse electrical readout
-> 2022: explicit receiver-relative phase and waveform differential (PWD)
-> 2022: projection through successively larger neural arrays (NAPOT)
-> 2024-2026: formalization, discriminators, and causal experiments
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The historical claim and the modern claim are related but distinct. The former supplies the intuition and mechanism ancestry. The latter supplies the typed variables, receiver reference, and failure conditions required for a scientific test.

4. A Morphology-Dependent Recognition Transform

Let the relatively slow learned state of source neuron (i) be

$$\mathbf{s}_i(t) = [\mathbf{m}_i, \mathbf{w}_i, \mathbf{r}_i, \mathbf{g}_i, \mathbf{c}_i], \quad (1)$$

where $\mathbf{m}_i$ describes branch morphology, $\mathbf{w}_i$ synapse placement and efficacy, $\mathbf{r}_i$ receptor state, $\mathbf{g}_i$ intrinsic conductances, and $\mathbf{c}_i$ other declared cellular organization. These components are not assumed to be statistically independent or equally stable.

Let $\mathbf{u}_i(t)$ be the current spatial-temporal input and $\mathbf{h}_i(t)$ the faster recent history. A morphology-sensitive match score is

$$q_i(t) = Q_i(\mathbf{u}_i(t), \mathbf{s}_i(t), \mathbf{h}_i(t)). \quad (2)$$

The resulting output is a typed vector,

$$\mathbf{y}_i(t) = F_i(q_i(t), \mathbf{s}_i(t), \mathbf{h}_i(t)), \quad (3)$$

with declared components such as event time, circular phase, instantaneous frequency, waveform height and duration, burst state, adaptation, and release-related measures. Equations 1-3 give a measurable interpretation of the 2012 statement that the neuron is a program: learned cellular state parameterizes the transformation performed on an arriving relation.

A structural feature counts as memory-bearing in this paper only if it improves prediction of a held-out input-output relation. Purely microscopic variation with no reproducible functional consequence is not counted as stored functional information.

5. From the Evoked Difference to a PWD

A receiver is never blank. It has an ongoing membrane state, oscillatory context, inhibitory condition, and expectation generated by recent activity. Let $\hat{\mathbf{y}}_{i,T}(t)$ denote the typed event expected under a declared tonic or receiver-relative reference T . For linear components,

$$\mathbf{d}_{i,L}(t) = \mathbf{W}_L [\mathbf{y}_{i,L}(t) - \hat{\mathbf{y}}_{i,L,T}(t)], \quad (4)$$

where $\mathbf{W}_L$ scales components into declared dimensionless units. Circular phase requires a circular residual,

$$d_{\phi,i}(t) = \text{Arg} \left\{ \exp \left(j[\phi_i(t) - \hat{\phi}_{i,T}(t)] \right) \right\}. \quad (5)$$

The complete typed departure is

$$\mathbf{d}_i(t) = [d_{\phi,i}, d_{\tau,i}, d_{f,i}, d_{A,i}, d_{D,i}, d_{B,i}, d_{R,i}]^\top, \quad (6)$$

where the subscripts denote phase, latency, frequency, amplitude, duration, burst, and release-related components. Only now, after the operation is defined, do we name $\mathbf{d}_i(t)$ a **Phase Wave Differential**. The term identifies a receiver-relative, typed change. It is not a new name for phase coding, a traveling wave, or predictive coding. Those established concepts can describe particular timing relations, propagating population patterns, or error-related architectures; PWD names the local difference created when an arriving event meets the current state of a particular receiving transform.

The morphology-specific effect is estimated by a controlled contrast,

$$\Delta_{\text{morph},i}(t) = \mathbf{d}_i(t | \mathbf{s}_i) - \mathbf{d}_i(t | \mathbf{s}_i^{(0)}), \quad (7)$$

where the reference state $\mathbf{s}_i^{(0)}$, input, recent history, and measurement protocol are declared.

6. What Is Transmitted

The source does not need to reproduce every morphological coordinate at the receiver. The testable transmission quantity is conditional information,

$$\mathcal{J}_{\text{morph}} = I(\mathbf{s}_i; \mathbf{z}_j \mid \mathbf{u}_i, \mathbf{h}_i, T_j), \quad (8)$$

where $\mathbf{z}_j$ is the measured response of receiver j and T_j is its declared tonic state. A receiver may recover a task-relevant class rather than a full source state,

$$\hat{k} = D(\mathbf{z}_j, \mathbf{u}_i, \mathbf{h}_i, T_j), \quad P(\hat{k} = k) > P_{\text{control}}. \quad (9)$$

This is the precise sense in which morphology is transmitted: learned structure constrains a functional response; the evoked response carries a sparse, typed consequence; and the receiver’s own structure determines what part of that consequence becomes causally available.

6.1 Incremental prediction and intervention criteria

The morphology term must earn its place by improving held-out prediction over a strong baseline. Let $\mathcal{L}_0$ be the loss of a model containing input, recent history, effective scalar synaptic strength, and rate, and let $\mathcal{L}_1$ add measured morphology and typed state. Define

$$\Delta_{\text{pred}} = \mathbb{E}_{\text{test}}[\mathcal{L}_0 - \mathcal{L}_1]. \quad (10)$$

The preregistered morphology prediction passes only when the confidence interval for Δ_{pred} excludes zero in the favorable direction across held-out cells or animals, not merely repeated trials from one cell.

For component r of the typed output, let $do(d_r \leftarrow d_r^{(0)})$ denote a selective perturbation toward baseline and $do(d_r \leftarrow d_r^*)$ a matched restoration. A causal receiver criterion is

$$\Delta_j^{(r)} = \mathbb{E}[z_j \mid do(d_r \leftarrow d_r^*)] - \mathbb{E}[z_j \mid do(d_r \leftarrow d_r^{(0)})]. \quad (11)$$

The direction of $\Delta_j^{(r)}$, the receiver population, and the rescue timing must be declared before intervention. This protects the proposal from a decoder that finds a correlation without identifying a route.

6.2 Preregistered claim ladder

Level	Required observation	Strongest warranted statement
A	morphology improves held-out prediction	measured structure contributes to the cellular transform
B	typed output distinguishes states at matched rate	information survives beyond scalar rate
C	receiver response depends on a typed component	the component is causally available downstream
D	loss and matched rescue follow the predicted route	morphology-to-PWD readout is supported for that route

No result at one level automatically licenses the next.

7. Deterministic Matched-Rate Demonstration

7.1 Purpose and design

The accompanying model asks a narrow identifiability question: can two cellular states be constructed so that they have identical event rate but remain distinguishable through phase, duration, burst, and downstream readout? The model generates 4,000 paired observations using fixed seed 20260808. Input pairs are split together between training and test sets so that paired-rate identity cannot leak across the split. Both state classes have the same scalar rate. They differ only in typed output relationships, and a downstream transform receives those typed relationships.

This is not a fit to neural data and is not used to estimate biological performance. Its function is to prevent a hidden mathematical contradiction: if a typed readout cannot outperform rate even in a controlled construction, the proposed discriminator would be ill-posed.

7.2 Results

Rate-only classification remained at chance, 0.500. The full typed PWD vector classified the two constructed states with accuracy 1.000, and the downstream readout also achieved 1.000. Ablating phase alone left accuracy at 1.000 because duration and burst structure were deliberately redundant in this fixture. Ablating every typed component returned accuracy to 0.500. Restoring the typed relation rescued accuracy to 1.000.

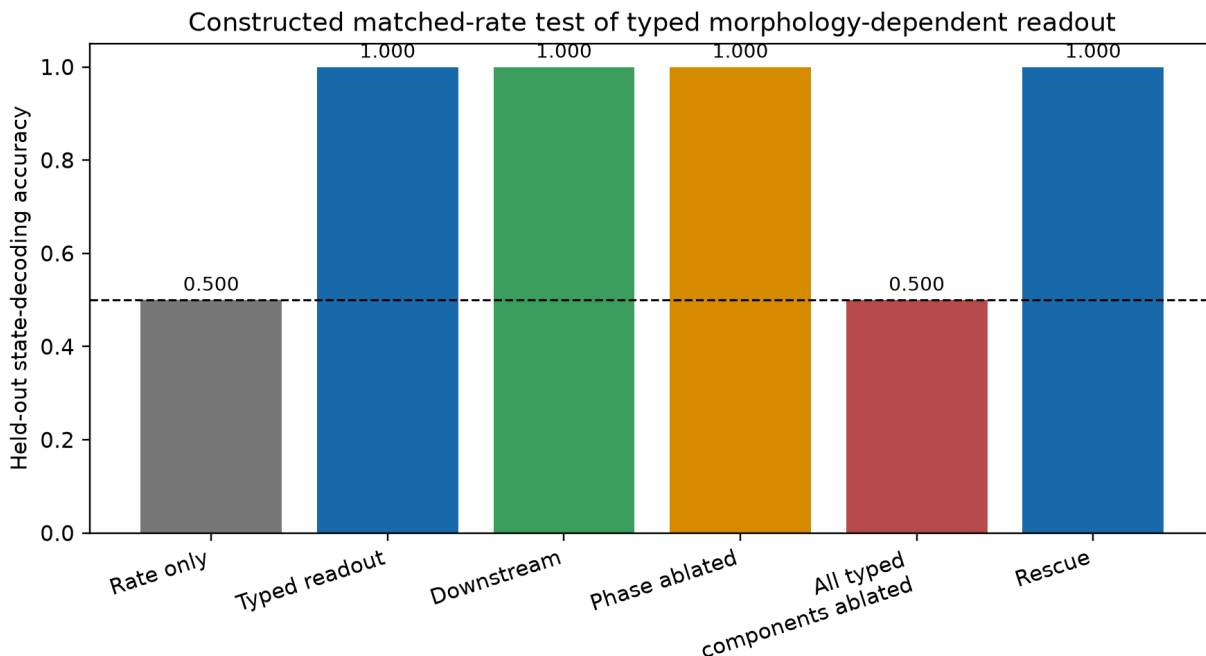


Figure 1: Matched-rate morphology-to-PWD demonstration

The phase-ablation result is important: this demonstration does not prove that phase is necessary. It shows that the proposed readout is typed and can contain redundant discriminating components. Biological necessity must be tested component by component.

All declared checks pass: exact rate pairing, a typed-over-rate margin greater than 0.30, return to chance after full typed ablation, and restoration after matched rescue. The result file records the model class, seed,

metrics, checks, artifact paths, and source-script SHA-256.

8. Decisive Experimental Program

8.1 Morphology-match-PWD pairing

Reconstruct identified dendritic arbors, map synapse locations and selected receptor/channel states, deliver controlled spatial-temporal input patterns, and record somatic and axonal outputs. Compare scalar-input and scalar-rate baselines with nonlinear-branch and full typed-output models on held-out trials. Structural variables support the proposal only if they improve out-of-sample prediction after total input, recent activity, and mean rate are controlled.

8.2 Causal branch intervention and rescue

Alter an identified branch feature, spine cluster, receptor distribution, or conductance while preserving the stimulation protocol. Predict in advance which match and output components should change. A strong test requires selective loss followed by matched restoration or an intervention that recreates the predicted functional relation through a different physical route.

8.3 Longitudinal learning

Measure the same cells before learning, during acquisition, after consolidation, and during recall. Test whether structural and molecular changes predict a stable change in the input-to-output transform. The critical comparison is not merely before versus after; it is whether the measured structural delta predicts the direction and magnitude of the typed output delta on held-out patterns.

8.4 Matched-rate receiver test

Construct two conditions with similar spike count and mean firing rate but different learned cellular states. Decode state from phase, latency, duration, waveform, burst, and release features. Then perturb one component while holding gross rate as constant as the preparation permits. The claim is strengthened only if identified receivers change selectively.

8.5 Source-receiver reference test

The same source event can have different effects at receivers in different phases or inhibitory states. Measure both ends of the route and vary receiver state independently. A receiver-relative PWD account predicts an interaction between the source event and receiver context, not a fixed meaning carried by the source alone.

9. Competing Accounts and Discriminators

Several established explanations can reproduce parts of the observation:

- a scalar synaptic-weight model may explain output once location and history are absorbed into effective weights;
- a nonlinear dendritic-subunit model may explain recognition without a distinct typed transmission claim;
- firing rate or spike timing alone may capture all receiver-relevant information;
- common input may create apparent source-receiver correspondence;

- adaptation may explain waveform differences without stable learned morphology; and
- a flexible decoder may exploit nuisance variables such as trial time, arousal, or movement.

The SAN account earns additional explanatory status only when morphology-sensitive state, receiver-relative typed difference, and causal downstream effect are all needed. A terminology comparison comes after the mechanism: **phase coding** describes information related to phase; **rate coding** describes information related to event rate; **traveling waves** describe spatial propagation; **predictive coding** describes families of inference and error architectures. PWD specifies the local, typed, source-and-receiver-relative difference that can be produced within any of those broader regimes. Its value depends on whether this added relational variable improves prediction and intervention.

10. Falsification and Revision Conditions

The central proposal is weakened or rejected if:

1. structural variables add no held-out predictive value beyond effective scalar input and recent history;
2. typed output differences vanish once mean rate and measurement artifacts are controlled;
3. identified receivers do not distinguish the typed outputs;
4. component-selective perturbations fail to change the predicted receiver response;
5. matched restoration fails to rescue that response; or
6. the same results are explained more parsimoniously by a declared comparator without the morphology-to-PWD link.

A failure of phase necessity would revise the phase component, not automatically erase duration, burst, waveform, or receiver-state contributions. Conversely, successful phase decoding alone would not establish the whole proposal.

11. Discussion

The strongest version of the 2012 insight is neither that a neuron literally broadcasts its anatomy nor that every microscopic detail is recoverable downstream. It is that a neuron’s learned material organization helps determine the function it performs, and that the function can leave a compressed, typed signature in its output. That signature can matter even when spike count does not change.

This reframes the neuron from a scalar relay into a stateful receive-transform-project system. Reception depends on learned sensitivity. Transformation depends on morphology, biophysics, recent history, and local context. Projection is an output whose effect is completed by a receiver. PWD names the measurable differential at that interface. The next paper in this sequence asks how much information a morphology could ideally, physically, and functionally support. The third asks how a local differential can be transformed and reinstated across larger arrays.

12. Conclusion

A learned neuron can be tested as a physical program without treating anatomy as software metaphor alone. If morphology and cellular state predict a pattern match, if the match produces a typed receiver-relative output beyond rate, and if selective perturbation and rescue control downstream responses, then a portion of structural memory has become causally available through a PWD readout. The proposal is specific enough to fail, and its failure conditions distinguish a genuine mechanism from a relabeling of established codes.

Research Ethics and Data Scope

This theoretical paper reports no new human or animal experiment. Any future invasive experiment described here requires appropriate institutional review, animal-welfare or human-subject protections, preregistration where feasible, and publication of exclusions and adverse events. All numerical results in the present manuscript come from constructed data generated by the included deterministic script.

Code, Data, and Reproducibility

The deterministic demonstration, fixed seed, machine-readable results, and figure are included with this manuscript:

- `model/run_morphology_pwd_demo.py`
- `model/results/morphology_pwd_demo_results.json`
- `model/results/morphology_pwd_demo_metrics.csv`
- `output/figures/morphology_pwd-matched-rate-demo.png`

The demonstration uses constructed data and makes no empirical effect-size claim.

Authorial Provenance and AI Assistance

The morphology-as-program and sparse electrical-readout ancestry is preserved in Micah Blumberg’s 2012 source record; the mature PWD and NAPOT terms arise in the 2022 SAN corpus. AI tools assisted source recovery, mathematical notation, model implementation, and editorial review. The historical source claims, theory program, and authorship belong to Micah Blumberg.

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